

K-12 Technology KPI Catalog

60+ CTO-Ready Measures by Leadership Domain

K-12 Technology KPI Catalog

Use this catalog as a starting point, not as a requirement to track everything. A good executive dashboard normally contains a small number of measures that answer current leadership questions.

GenAI data guardrail: Use fictional, public, aggregated, or properly sanitized information with public generative AI tools. Do not enter student records, staff records, personally identifiable information, credentials, sensitive cybersecurity details, legal records, confidential procurement material, vendor-confidential information, or sensitive district project information unless your district has explicitly approved the tool and data use.

Project Management

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Milestones completed on time	% of due milestones completed by planned date	On-time due milestones / all due milestones x 100	Higher	Biweekly	Project sponsor / cabinet
Schedule variance	Difference between planned and actual milestone timing	Actual days - planned days	Toward 0	Biweekly	Project lead
Budget variance	Difference between approved/forecast spending and actual/committed spending	Actual or committed - planned	Toward 0	Monthly	CTO / CFO
Open high risks	Unresolved risks rated high for probability and impact	Count	Lower	Weekly	Project sponsor

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Scope changes approved	Approved additions/removals that change baseline scope	Count and estimated impact	Context	Monthly	Project sponsor
Decisions overdue	Leadership decisions past the date needed to protect schedule	Count	Lower	Weekly	Cabinet / sponsor
Benefits realized	% of defined benefits meeting agreed evidence criteria	Benefits achieved / benefits due x 100	Higher	Quarterly	Sponsor / cabinet
Stakeholder confidence pulse	Favorable response to a short project confidence question	Favorable responses / all responses x 100	Higher	Monthly	Project lead

Infrastructure and Reliability

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Core network availability	% of scheduled instructional time core network services are available	Available minutes / scheduled minutes x 100	Higher	Monthly	CTO / cabinet
Internet edge availability	% of scheduled time internet service is available	Available minutes / scheduled minutes x 100	Higher	Monthly	CTO / cabinet
Critical application availability	% of scheduled time selected district services are usable	Available minutes / scheduled minutes x 100	Higher	Monthly	IT leadership
Wireless service-impact incidents	Incidents causing meaningful classroom wireless interruption	Count	Lower	Monthly	Infrastructure team
Capacity threshold exceptions	Sites/services exceeding locally defined utilization threshold	Count of exceptions	Lower	Monthly	Infrastructure team

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Backup recovery tests passed	% of scheduled restore tests completed successfully	Successful tests / planned tests x 100	Higher	Quarterly	CTO / risk team
UPS lifecycle compliance	% of identified UPS batteries within district replacement standard	Compliant units / tracked units x 100	Higher	Quarterly	Infrastructure / budget
Infrastructure projects on lifecycle plan	% of major assets mapped to replacement year and funding path	Assets with plan / assets in scope x 100	Higher	Quarterly	CTO / CFO

Help Desk and Service

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Median first response	Median business time from ticket creation to first meaningful response	Median elapsed business time	Lower	Weekly	Help Desk lead
Median resolution time	Median business time from ticket creation to resolution	Median elapsed business time	Lower	Weekly	Help Desk lead
Tickets resolved within SLA	% of tickets resolved within priority-based service target	SLA-compliant tickets / eligible tickets x 100	Higher	Weekly	CTO / principals
Backlog aging	% of open tickets older than locally defined threshold	Aged open tickets / all open tickets x 100	Lower	Weekly	Help Desk lead
First-contact resolution	% of eligible tickets resolved without escalation or repeat contact	First-contact resolutions / eligible tickets x 100	Higher	Monthly	Support team
Repeat incident rate	% of resolved tickets reopened or repeated for same issue within window	Repeat incidents / resolved tickets x 100	Lower	Monthly	Support team
Customer satisfaction	Favorable support survey responses	Favorable / responses x 100	Higher	Monthly	CTO / principals

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Instructional-impact ticket share	% of tickets classified as directly interrupting teaching/learning	Instructional-impact / all tickets x 100	Context	Monthly	CTO / curriculum

Devices and Asset Management

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Device readiness	% of assigned instructional devices ready for intended use	Ready devices / assigned devices x 100	Higher	Monthly	CTO / principals
Inventory accuracy	% of sampled assets matching inventory records	Matching assets / sampled assets x 100	Higher	Quarterly	CTO / finance
Devices beyond lifecycle	% of active devices older than district replacement standard	Over-age devices / active devices x 100	Lower	Quarterly	CTO / CFO
Warranty coverage	% of active devices still covered by intended warranty/support plan	Covered devices / in-scope devices x 100	Context	Quarterly	Operations
Repair turnaround	Median business time from repair intake to return-to-service	Median elapsed business time	Lower	Monthly	Operations
Loaner availability	% of campuses meeting district loaner reserve standard	Campuses meeting standard / campuses x 100	Higher	Monthly	Operations / principals
Deployment milestone completion	% of planned deployment quantity completed by date	Completed / planned x 100	Higher	Weekly during rollout	Project sponsor
Asset disposition closure	% of retired assets documented through approved disposal process	Closed dispositions / retired assets x 100	Higher	Quarterly	CTO / finance

Cybersecurity and Privacy

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
MFA coverage	% of eligible accounts protected by district-required MFA	$\frac{\text{MFA-enabled eligible accounts}}{\text{eligible accounts}} \times 100$	Higher	Monthly	CTO / cabinet
Required awareness training completion	% of assigned staff completing required security training by deadline	$\frac{\text{Completed}}{\text{assigned}} \times 100$	Higher	Monthly	Cybersecurity lead
Critical vulnerability remediation within standard	% of tracked critical findings remediated within district timeline	$\frac{\text{Remediated on time}}{\text{due findings}} \times 100$	Higher	Monthly	Security leadership
Privileged account review completion	% of scheduled privileged-access reviews completed	$\frac{\text{Completed reviews}}{\text{planned reviews}} \times 100$	Higher	Quarterly	CTO / security
Vendor reviews completed before approval	% of in-scope tools receiving required privacy/security review before use	$\frac{\text{Reviewed before approval}}{\text{in-scope approvals}} \times 100$	Higher	Monthly	Privacy / procurement
Backup restore assurance	% of required recovery tests passing documented acceptance criteria	$\frac{\text{Passing tests}}{\text{tests performed}} \times 100$	Higher	Quarterly	CTO / risk team
Incident response exercise completion	% of planned exercises completed with after-action review	$\frac{\text{Completed exercises}}{\text{planned exercises}} \times 100$	Higher	Quarterly	CTO / cabinet
Security exceptions overdue	Approved temporary exceptions past review/expiration date	Count	Lower	Monthly	CTO / risk team

Budget, Procurement, and Vendors

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Budget utilization	% of annual technology budget spent or committed relative to time/plan	$\text{Spent} + \text{committed} / \text{approved budget} \times 100$	Target range	Monthly	CTO / CFO
Forecast variance	Difference between year-end forecast and approved budget	$\text{Forecast} - \text{approved}$	Toward 0	Monthly	CTO / CFO
Renewals reviewed before deadline	% of renewals receiving value/privacy/security review before decision date	$\text{Reviewed on time} / \text{due renewals} \times 100$	Higher	Monthly	CTO / procurement
Unused license rate	% of paid licenses not meeting local activity threshold	$\text{Inactive paid licenses} / \text{paid licenses} \times 100$	Lower	Quarterly	CTO / business office
Procurement cycle time	Median time from complete request to award/PO for defined purchase types	Median elapsed business days	Lower	Quarterly	CTO / procurement
Lifecycle plan funded	% of forecast replacement need covered by identified funding	$\text{Funded replacement amount} / \text{planned replacement need} \times 100$	Higher	Annual/quarterly	CTO / CFO

Change Adoption and Professional Learning

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Awareness confirmation	% of affected users indicating they understand why the change is occurring	$\text{Favorable pulse} / \text{responses} \times 100$	Higher	At launch	Project/change lead
Required training completion	% of affected users completing required learning before milestone	$\text{Completed} / \text{assigned} \times 100$	Higher	Weekly during rollout	PD/change lead

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Activation rate	% of intended users completing first required setup/action	Activated / intended users x 100	Higher	Weekly	Project lead
Meaningful active use	% of intended users meeting a defined use threshold	Users meeting threshold / intended users x 100	Higher	Monthly	Project sponsor
Proficiency check	% of sampled users completing critical workflow without assistance	Successful / sampled x 100	Higher	Post-training	Change lead
Post-launch support demand	Support requests per 100 affected users	Support contacts / affected users x 100	Lower after stabilization	Weekly	Support/change lead
Champion coverage	% of campuses/ departments with an active local champion	Units with champion / units in scope x 100	Higher	Monthly	Change lead
Sustained adoption	% of intended users still meeting use expectation after defined reinforcement period	Sustained users / intended users x 100	Higher	60/90 days	Sponsor

Software, Cloud, and Data Services

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
SSO coverage	% of approved high-use applications integrated with district SSO where appropriate	SSO-integrated / in-scope apps x 100	Higher	Quarterly	Identity lead
Automated account provisioning	% of in-scope account lifecycle events handled automatically	Automated events / in-scope events x 100	Higher	Monthly	Identity/SIS
Data integration success	% of scheduled critical data jobs completing within acceptance window	Successful jobs / planned jobs x 100	Higher	Weekly	Data systems

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Approved application inventory completeness	% of known applications represented in governance inventory	Documented apps / known apps x 100	Higher	Quarterly	CTO / privacy
Platform adoption	% of intended users meeting locally defined meaningful-use threshold	Active users / intended users x 100	Higher	Monthly	CTO / curriculum
De-implementation savings realized	% of approved savings from retired tools/services actually removed from spend	Realized savings / approved savings x 100	Higher	Quarterly	CTO / CFO

Communication and Customer Experience

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Status update cadence met	% of planned project/ service updates published on schedule	Updates on time / updates planned x 100	Higher	Monthly	Project/ communications lead
Stakeholder message reach	% of intended audience receiving/opening a required communication where measurable	Reached / intended x 100	Higher	Per campaign	Communications
FAQ deflection	% of common inquiries resolved through published self-service resource	Self-service resolutions / related contacts x 100	Higher	Monthly	Support/ communications
Principal confidence	Favorable principal response to technology service pulse	Favorable / responses x 100	Higher	Quarterly	CTO
Service transparency	% of high-impact incidents/ projects with current public/internal status update as required	Current updates / in-scope items x 100	Higher	Monthly	CTO / communications

KPI	What It Measures	Example Formula	Direction	Cadence	Typical Audience
Board actionability rating	Internal review of whether dashboard items clearly identify status, trend, and action/decision	Items meeting standard / items reported x 100	Higher	Per board cycle	CTO / superintendent

How to Use the Catalog

1. Start with the decision or leadership question.
2. Select the smallest set of KPIs that answers it.
3. Define the formula and data source locally.
4. Establish a baseline before setting an ambitious target.
5. Name an owner for the measure and an owner for the response.
6. Decide what action is triggered when the KPI is off target.
7. Review whether the KPI is still useful after the project or service stabilizes.

Source and Alignment Notes

This sample library is designed as a companion to **Module 8: Project Management and Change Leadership** in the TCEA Technology/IT Director Certification. It extends the module's emphasis on desired outcomes, progress reporting, KPIs, dashboards, communication, change adoption, and responsible generative AI use.

Useful reference points:

- Make Data Displays Without Writing Code - TCEA TechNotes article on GenAI-assisted data displays, dashboard workflows, Google Sheets, GitHub Pages, and embedding.
- KPI Dashboards Built with Gen AI - TCEA TechNotes article demonstrating a browser-based dashboard paired with an editable Markdown data file.
- Metrics and KPIs Made Easy - TCEA TechNotes introduction to actionable KPI scorecards for technology leaders.
- K-12 KPI Dashboard Generator - browser-based KPI planning and dashboard starter.
- District KPI Dashboard Example - example of a scoreboard-style district dashboard.
- Data Display Patterns Field Guide - examples, prompts, and hosting considerations for lightweight visual displays.

Data note: Every Ardent ISD number in this library is illustrative sample data. Replace it with locally verified data before operational use.